MILITARY SPECIFICATION

KETTLES, STEAM JACKETED; TWIN; 5-GALLON, TILTING,

CABINET MOUNTED; NATURAL GAS HEATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two 5-gallon capacity, tilting, steam jacketed kettles, cabinet mounted, and a self-contained gas heated, steam generator boiler.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-T-60 - Tape: Packaging, Waterproof

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116 - Preservation-Packaging, Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

FSC 7310

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Gas Association, Inc. (A.G.A.)

Directory of Certified Appliances and Accessories

(Application for copies should be addressed to the American Gas Association, Inc., 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

American National Standards Institute (ANSI)

ANSI - Z21.46 - Gas-Fired Kettles, Steam Cookers, and Steam Generators

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018, or the American Gas Association, 8501 East Pleasant Valley Road, Cleveland, OH 44131.)

American Society of Mechanical Engineers (ASME)

ASME Boiler and Pressure Vessel Code Titled

Section IV - Rules for Construction of Heating Boilers
Section VIII - Unfired Pressure Vessel Code

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

American Society for Testing and Materials (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip

A 240 - Stainless and Heat-Resisting Chromium and Chromium Steel Plate, Sheet and Strip for Fusion-Welded Unfired Pressure Vessels

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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National Sanitation Foundation (NSF)

Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment

NSF Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

Underwriters' Laboratories, Inc. (UL)

UL 197 - Commercial Electric Cooking Appliances

(Application for copies of UL publications should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 1285 Walt Whitman Road, Melville, L.I., NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The tilting kettle assembly delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with requirements specified herein. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

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3.2 Codes and standards. The tilting kettle assembly shall comply with the requirements of UL 197, ANSI Z21.46; the ASME Boiler and Pressure Vessel Code, Sections IV and VIII; and NSF Standard No. 4, as applicable.

3.2.1 Compliance. Prior to the approval of the initial production lot, the supplier shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the tilting kettle assembly he proposes to furnish under this specification meets the applicable requirements of UL 197, ANSI Z21.46; ASME Boiler and Pressure Vessel Code, Sections IV and VIII, and NSF Standard No. 4.

3.2.1.1 ANSI. Acceptable evidence of meeting the requirements of ANSI Z21.46 may be a photostatic copy of the A.G.A. Appliance Certificates; a listing of the tilting kettle assembly in the A.G.A. Directory of Certified Appliances and Accessories; or a certified test report from a recognized, independent testing laboratory acceptable to the Government stating that the tilting kettle assembly has been tested and conforms to ANSI Z21.46.

3.2.1.2 ASME code. Acceptable evidence of meeting the applicable requirements of the ASME Boiler and Pressure Vessel Code, Section IV and VIII shall be the ASME symbol stamped on the boiler and kettles, respectively.

3.2.1.3 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished tilting kettle assembly, or

(2) A certification for the tilting kettle assembly issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the tilting kettle assembly is being procured indicating that the tilting kettle assembly has been tested and conforms to NSF Standard No. 4.

3.2.1.4 UL. Acceptable evidence of meeting the applicable requirements of UL 197 shall be the UL label, the UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the tilting kettle assembly offered has been tested and conforms to the applicable requirements of UL 197.

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3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial tilting kettle assembly, provided the finished item complies with all provisions of this specification.

3.3.1 Stainless steel.

3.3.1.1 Plate, sheet, and strip. Stainless steel, plate, sheet, and strip shall be type 302 or 304 of ASTM A 167.

3.4 Design and construction. The tilting kettle assembly shall consist of two 5-gallon capacity stainless steel steam jacketed tilting kettles with trunnion(s) mounted on an enclosed cabinet base. The cabinet base shall be provided with two doors and shall house a self-contained ASME approved steam generator boiler, piping, gage, and controls necessary for safe and efficient operation. The steam generating boiler shall be gas heated. The completely assembled unit, i.e. cabinet base with steam generating boiler and tilting kettles, shall be ready for connection to drain, water, gas and electrical supplies. The tilting kettles shall be designed to cook, simmer, or keep food products warm in a temperature range of 150 Fahrenheit (F) to not less than 240 F by means of steam generated in the steam generating boiler. Dimensions and connection sizes for the tilting kettle assembly and steam generating boiler shall conform to table I. All kettles, trunnion mounting(s), steam generator boiler, piping, joints and fittings shall be assembled to provide leakproof components and joints (see 4.4.1).

TABLE I. Requirements

Requirements	Dimensions and connection sizes
Overall outside diameter at kettle rim (max)	23 inches
Height from bottom of kettle mount to kettle rim (max)	23 inches
Clearance between kettle spout (in full tilt position) and top of stand (min)	4-1/2 inches
Size of hot and cold water inlet (nominal tubing size)	3/8 inch
Size of gas connection	1 inch (IPS) maximum
Size of drain connection	2 inches (IPS) maximum

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TABLE I. Requirements (cont'd)

Requirements	Dimensions and connection sizes
Size of safety valve connection	3/4 inch (IPS)
Clearance between top of kettle and swing spout outlet	1-1/2 inches ± 1/4 inch
Overall dimensions kettles and cabinet base (min)	Width - 36 inches Depth - 33 inches Height - 54 inches

3.4.1 Tilting kettles. Tilting kettles shall be designed in accordance with Section VIII of the ASME Boiler and Pressure Vessel Code and shall have a design pressure not less than 30 pounds per square inch gage (psig). The tilting kettles shall be thermostatically operated at any steam jacket pressure up to 15 psig. The tilting kettles shall consist of a kettle body, steam jacket, tilting handle, trunnion mounting(s), safety valve and plumbing.

3.4.1.1 Kettle body. The body shall be cylindrical with a hemispherical or sloping bottom and shall be fabricated of stainless steel as specified in 3.3.1.1. Material thickness of that section of the body forming the inner wall of the steam jacket shall conform to the requirements of ASME for the rated working pressure of the kettle. The body shall be seamless drawn or spun from one piece of material. The top edge shall be formed into a minimum 5/16 inch, half round, open radius, a fully closed sealed bead, or a double hem. A V-shaped pouring lip either formed as a part of the body or welded to the top edge shall be provided.

3.4.1.2 Steam jacket. The steam jacket shall be stainless steel. Material thickness shall conform to the requirements of ASME for the rated working pressure of the kettle. The jacket shall be formed or spun from one piece or shall be fabricated by welding preformed segments. The top of the jacket shall be welded to the kettle body. A means for complete drainage of the water jacket shall be provided.

3.4.1.3 Tilting handle. The tilting handle shall consist of a stainless steel rod and a heat resistant plastic knob. The handle shall be positioned on the outboard side of each kettle when facing the kettle front, and the rod shall be inserted into the trunnion shaft or into a lug welded to the kettle body above the steam-water jacket. The knob shall be a push-on type or be threaded to receive the threaded end of the rod.

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3.4.1.4 Kettle mountings. Kettles shall be provided with either one or two trunnions welded to the kettle body. When one trunnion is provided, the kettle shall be mounted in a combination kettle support and tilting trunnion. The combination support shall be fabricated of stainless steel or cast aluminum finished as specified in 3.8. When two trunnions are provided, the trunnion shafts shall be welded to the body above the steam-water jacket and the trunnion bushings shall be mounted on two stainless steel leg supports. The kettle provided with a combination support shall be equipped with a stainless steel base plate. Both types of kettle mountings shall have provisions for bolting the kettle assembly to the cabinet specified in 3.4.2. The trunnion(s) shall allow 95 degrees of tilt from the normal position and shall allow the kettle to return to the upright position when released. The trunnions and supports shall show no visible signs of deformation when tested as specified in 4.4.4.

3.4.1.5 Cover. Each kettle shall be provided with a one-piece removable cover, formed from minimum 0.056 inch stainless steel specified in 3.3.1.1. The cover shall be flanged, dished, or suitably reinforced to prevent distortion as a result of normal use and shall be provided with a knob or loop-bar type handle. The top of the kettle body shall fit the flange on the cover.

3.4.2 Cabinet base. The cabinet base shall be of the totally enclosed type enclosing a steam generator boiler and shall consist of a stainless steel base, marine edged cabinet top, two hinged doors, and four adjustable legs. The cabinet top shall be provided with provisions for mounting each tilting kettle, trunnion(s), single swing spout faucet with shut-offs (hot and cold water), and steam valves. The counter top shall include, for each tilting kettle, a sink with drain, removable grate, and removable drip tray. The cabinet base shall contain a galvanized or standard, welded, angle-iron frame enclosed by an outer skin of not less than 0.0625 inch (16 gage) thick stainless steel specified in 3.3.1.1. Each cabinet door shall be hinged and supplied with a magnetic latch and a handle. The cabinet doors shall provide access to the steam generator boiler, controls, valves, and plumbing. The cabinet doors shall be fabricated from not less than 0.0625 inch (16 gage) stainless steel specified in 3.3.1.1. The cabinet top shall be fabricated from not less than 0.0500 inch (18 gage) thick stainless steel specified in 3.3.1.1.

3.4.2.1 Steam generator boiler. The steam generator boiler shall be designed in accordance with Section IV of the ASME Boiler and Pressure Vessel Code for operation at pressures not exceeding 15 psig.

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3.4.2.1.1 Safety valve. Each kettle shall be provided with a pressure-type, brass, safety valve constructed in accordance with applicable requirements of ASME. The valve shall be set at the factory in accordance with the ASME Section VIII Code for the maximum working pressure of the kettle. The valve shall be furnished with a reversible lifting level and shall be capable of being rotated 360 degrees without disturbing the valve settings when tested as specified in 4.4.1. The valve shall be provided with a bottom inlet and side discharge outlet and shall be connected to the steam jacket by means of a shoulder-nipple and elbow.

3.4.2.2 Legs. The cabinet shall be mounted on four, six inch high stainless steel legs with adjustable bullet feet. Foot adjustment shall be not less than 1/2 inch. Legs may be either removable or permanently attached by welding to the frame.

3.4.2.3 Controls. Each kettle shall have a manually adjustable steam inlet control valve to control the steam supply from the boiler to the kettle. The control valve(s) shall operate without binding and shall show no evidence of steam leakage when tested as specified in 4.4.1.

3.4.2.4 Pipe. Pipe shall be Iron Pipe Size (IPS) with American Standard taper pipe thread (NPT) ends. Pipe for water lines shall be seamless brass, standard weight.

3.4.2.5 Pipe fittings. Pipe fittings used with brass pipe shall be bronze, and of the type and weight normally furnished by the manufacturer on his standard product. Pipe fittings welded to the steam-water jacket shall be stainless steel when the jacket is stainless steel.

3.4.2.6 Globe valve(s). Globe valve(s) shall be rated for hot or cold water pressures up to 125 psig. Valve(s) shall be of composition bronze and shall have a removable disc-holder with removable discs suitable for water temperatures up to 212°F. Valve bodies shall have threaded ends. The working pressure shall be cast on the side of the valve body as acceptable evidence of compliance with pressure requirements.

3.4.2.7 Pressure gage. The pressure gage shall be of the dial-type and shall be graduated in pounds from 0 to 30 psig minimum and numbered at 15 pounds or less. Dial face shall be not less than 2 inches in diameter. The pressure gage shall conform to the applicable requirements of ASME code specified in 3.4.2.1. The pressure gage shall be located so as to be visible to the operator. When tested as specified in 4.4.1, pressure gage reading shall not vary more than plus or minus 1 psi from actual steam generator boiler pressure.

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3.4.2.8 Swing-spout. A swing-spout water supply connection shall be provided. The spout shall be a combination swing-spout and faucet or a swing-spout with globe valves. The swing-spout shall be the manufacturer's standard design and shall be chrome-nickel-plated brass or stainless steel. The swing-spout shall be mounted on the cabinet between each of the 5-gallon tilting kettles. When a faucet is not an integral part of the swing-spout, globe valves (see 3.4.2.6) of the same size as the water connections (hot and cold) shall be installed in the plumbing directly below the swing-spout. The spout outlet shall be provided with an aerator.

3.4.2.9 Automatic drain unit. When the gas steam generator boiler is provided with an automatic drain unit. Automatic drain units shall be automatically filled and brought to pressure whenever heat and water controls are turned on and while under pressure and shall automatically drain whenever heat and water switches are turned off when tested as specified in 4.4.1. A cold water condenser shall be provided. The cold water condenser shall reduce the temperature of water entering the drain line.

3.4.3 Gas characteristics. The gas steam generator boiler shall operate on natural gas. Burner gas valves and pilot gas valves shall be of the universal or fixed orifice type. Valves of the fixed orifice type shall be provided with interchangeable orifice hoods or bushings drilled for natural gas. The valve hoods for natural gas shall be unpainted brass. The boiler shall have a minimum input of 170,000 British thermal units (Btu) (see 4.4.1).

3.4.3.1 Pressure regulator. A pressure regulator shall be furnished for use with natural gas fuel and the incoming water column gas pressure shall be between 3-1/2 - 7 inches for natural gas.

3.4.3.2 Automatic ignition. Burners shall be equipped with a means for automatic ignition of gas. A 100 percent shut-off device is required for all automatic ignition of gas and for all burners. The safety shut-off device shall automatically shut off the gas to all burners including pilot burners in event that means of ignition become inoperative (see 4.4.1).

3.4.4 Electrical characteristics. Unless otherwise specified (see 6.2) the electrical components for the tilting kettle assembly shall be suitable for operation on a nominal 120 volt 60 Hertz (hz) system.

3.5 Repair and maintenance. Provisions shall be made to allow access to inspect, service, replace, and adjust any component of the tilting kettles and steam generator boiler. Adjustment, repair, or replacement of any component shall not require disassembly or removal of additional component(s).

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3.6 Marking. Identification and operating instructions shall be permanently marked directly on the item or on an aluminum, brass, or stainless steel plate permanently affixed to the item. Marking shall be stamped embossed, engraved or applied by photosensitive means. The plate thickness shall be not less than 0.012 inch.

3.6.1 Identification marking. Each steam-jacketed tilting kettle assembly identification plate shall include the manufacturer's model and serial number and the manufacturer's name, trade name or trademark of such known character as to be readily identifiable to the manufacturer. In addition, such information as required by ANSI Z21.46 shall also be on the plate.

3.6.2 Users instruction plate. The plate shall be of sufficient overall size to contain, in legible characters, the following information. The words "STEAM-JACKETED KETTLE" shall be not less than 3/16 inch high and the words "USER'S INSTRUCTIONS" shall be not less than 1/8 inch high. All other characters shall be not less than 3/32 inch high. The plate shall be mounted on the inner door of the cabinet.

STEAM-JACKETED KETTLE

USER'S INSTRUCTIONS

1. Caution: Steam supply shall not exceed 15 psig.
2. Do not tamper with or obstruct safety valve.
3. Do not clean with abrasives.
4. Wash kettle before and after each use with hot soapy water and rinse thoroughly with clean hot water.
5. Do not turn steam on unless water or food is in the kettle.
6. Do not put water in hot dry kettle.
7. When heating a cold kettle, turn the steam on gradually, thus allowing the kettle to become warm before full pressure is applied. After kettle becomes warm, and before applying full pressure, open the safety valve momentarily to release any air trapped within the jacket.

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3.7 Welding. The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, or other foreign matter. Welds shall be sound and free from porosity, cracks or incomplete fusion. Exposed welds shall be free from burrs or sharp projections. Welding shall be made with welding rods having the same chemical composition as the welded material. A certificate of compliance verifying compatibility of the welding rod composition to the base material shall be furnished (see 4.3). The finished weld shall be smooth and uniform.

3.8 Finish. The tilting kettle assembly finish shall conform to NSF Standard No. 4 and as follows: Exposed piping and fittings, hinges, and latches when not stainless steel or aluminum shall be chrome-plated in accordance with the manufacturer's standard practice. Standard welded angle iron frame when provided, shall have a rustproof painted finish.

3.9 Workmanship. Components shall be free of all slivers, burrs, and rough or sharp edges. Bolt holes and rivet holes shall be accurately punched or drilled. Washers or lockwashers shall be provided in accordance with standard commercial practice, and all bolts, nuts, and screws shall be drawn tight. Rivets shall be tightly driven and fully peened.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.2.2 End item inspection. The inspection lot shall consist of all tilting kettle assemblies offered for inspection at one time. The sample unit shall be one completely fabricated and assembled tilting kettle assembly.

4.2.2.1 Visual examination. The tilting kettle assembly shall be examined for defects classified in table II. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Design, construction and workmanship (general)	Not finished in accordance with NSF Standard No. 4	X	
	Specified components not chrome-plated		X
	Fractured, dented, or bowed component	X	
	Rough or sharp edges, burrs, or slivers	X	
	Component not readily accessible for servicing where required		X
	Standard welded angle iron finish not rustproof	X	
Casting (where required)	Cracked, porous, blowholes, or other casting imperfections	X	
Welding (where required)	Incomplete, burn holes, cracked, fractured, or not fused	X	
	Slag inclusion, slight undercut, scale or flux deposits not removed		X
	Weld not smooth or non-uniform cross-section	X	
Bolts, nuts, screws, studs and other types of threaded fasteners	Missing, broken, stripped, or fractured	X	
	Loose		X
	Not equipped with washers or lockwashers		X
Rivets	Loose; head not in full contact with member		X

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Kettle	Body bottom not hemispherical or sloping	X	
	Not equipped with V-shaped pouring lip	X	
Jacket	Not equipped with means for draining	X	
Tilting handle	Not positioned on outboard side of each kettle	X	
	Not located above steam-water jacket	X	
	Not equipped with heat resistant knob	X	
Mount	One or two trunnions not properly welded to kettle body	X	
	Not bolted to cabinet base	X	
	Will not allow 95 degrees of kettle movement	X	
Cover	Not one piece construction	X	
	Not removable type	X	
	Not as specified, i.e. flanged	X	
	Missing	X	
	Handle not knob or loop-bar type	X	
	Handle loose		X
Cabinet base	Adjustable legs missing or not as specified	X	
	Frame not angle-iron	X	
	Top not provided with sink, drains, removable grates, removable drip-trays, trunnion mounting(s), swing-spout, or faucet	X	
	Top not marine edged type	X	
Generator	Not equipped with safety valve	X	

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TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Doors	Not as specified (stainless steel)	X	
	Not provided with hinged doors	X	
	Magnetic latches missing	X	
	Door does not provide access to boiler	X	
	Handles missing	X	
Controls	Missing or not as specified	X	
	Controls not accessible for servicing	X	
	Piping and valves not concealed in cabinet base	X	
	Valve working pressure not cast on side		X
Pressure gage	Missing	X	
	Dial not graduated as specified	X	
	Not visible to operator	X	
Marking for identification	Missing, incomplete, not legible, not affixed in the prescribed manner	X	
Marking instructions	Missing, incomplete, not legible, not specified type or size, not affixed in the prescribed manner	X	

4.2.2.2 Dimensional examination. The tilting kettle assembly shall be examined for compliance with dimensions specified. Any dimension not within the specification tolerance shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.2.3 End item testing.

4.2.2.3.1 Initial production lot. One tilting kettle assembly from the initial production lot shall be tested as specified in 4.4.1 through 4.4.4. Failure to pass any test shall be cause for rejection of the lot.

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4.2.2.3.2 Production lots. Each tilting kettle assembly shall be tested in accordance with 4.4.1 and 4.4.2. Failure to pass either test shall be cause for rejection of the lot.

4.2.2.4 Packaging inspection. An inspection shall be made to determine that packaging, packing, and marking complies with the section 5 requirements. Defects shall be as specified in table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

TABLE III. Inspection of packaging

Examine	Defect
Marking	Missing; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Components missing, damaged, or otherwise defective
Workmanship	Inadequate application of components, such as: preservation missing or not dried, surfaces not cleaned, openings not sealed, crate not anchored, blocked and braced as specified Crate bulged or distorted

4.2.2.4.1 Examination of shipping container. When shipping containers are required to be in accordance with PPP-B-601, inspection for defects in closure and strapping shall be in accordance with the appendix of the container specification.

4.3 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Tests.

4.4.1 Operational test. The tilting kettle assembly shall be given an operational test for not less than 15 minutes to determine that controls and valves operate as specified in 3.4.2.3 and 3.4.3.2 and that there is no

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evidence of leakage as specified in 3.4. Safety valve functions in accordance with 3.4.2.1.1. During this test, the temperature shall be checked for conformance with 3.4 and the Btu input of the steam generating boiler shall be checked for conformance with 3.4.3. Pressure gage accuracy shall be tested at 30 psig to determine conformance with 3.4.2.7. Automatic drain unit shall function in accordance with 3.4.2.9. Any nonconformance with requirements specified shall constitute failure of this test.

4.4.2 Hydrostatic test. Each tilting kettle and steam generator boiler shall be tested in accordance with the applicable section of the ASME Boiler and Pressure Code at the hydrostatic pressure specified herein. Any unit that fails to pass this test shall be rejected. Certification in accordance with 3.2.1.2 will be acceptable evidence that the unit has satisfactorily passed this test.

4.4.3 Capacity. Each tilting kettle shall be tested to determine compliance with capacity requirements of 3.4. The test shall be performed by filling the kettle with water from a container of known capacity. Any noncompliance with specified requirements shall constitute failure of the test.

4.4.4 Trunnions, support(s). Each tilting kettle shall be loaded with weights equivalent to 1-1/2 times the weight of its water capacity and the trunnion support(s) shall be checked to determine conformance with the deformation requirements of 3.4.1.4. Any nonconformance with the requirements specified shall constitute failure of this test.

5. PACKAGING

5.1 Preservation - packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. All exposed, uncoated, ferrous metal surfaces except corrosion-resisting steel of each cooker shall be cleaned process C-1 of MIL-P-116 and thoroughly dried.

5.1.1.2 Application of preservation. Surfaces clean as specified shall be coated with type P-14 preservative of MIL-P-116.

5.1.1.3 Sealing of openings. All openings, such as inlets, outlets, vents, valves, and the face of the steam gage shall be sealed with tape conforming to type III, class 1, or type IV of PPP-T-60.

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5.1.1.4 Doors. The doors of the cabinet base shall be securely closed so as to prevent their opening while in transit.

5.1.2 Level C. Each tilting kettle assembly shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A packing. Each tilting kettle assembly preserved and packaged as specified in 5.1, shall be packed in a plywood crate conforming to type I, class 2, style a of MIL-C-104. Anchoring, blocking and bracing shall be in accordance with the appendix of MIL-C-104.

5.2.2 Level B packing. Each tilting kettle assembly preserved and packaged as specified in 5.1, shall be packed in a cleated plywood shipping container conforming to domestic type, style A or B of PPP-B-601. The tilting kettle assembly shall be secured with lumber blocking and bracing and a minimum of 1 inch shall be maintained between the legs of the tilting kettle assembly and the inside surface of the bottom of the shipping container.

5.2.3 Level C packing. Tilting kettle assemblies, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tilting kettle assembly covered by this specification is intended for cooking of food in kitchens of enlisted personnel dining facilities.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Electrical requirements, if other than specified (see 3.4.4).
(c) Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).

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6.3 Contract data. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity included in DD Form 1423, Contract Data Requirements List, and cited in the contract or purchase order.

Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity:

Army - GL

Project No. 7310-A543

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE KETTLES, STEAM JACKETED; TWIN; 5-Gallon, TILTING, MIL-K-43943(GL) CABINET MOUNTED; NATURAL GAS HEATED

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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